

Experiment 1

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1. Aim:

To design and implement a sample database system using DDL, DML, and DCL commands, including database creation, data manipulation, schema modification, and role-based access control to ensure data integrity and secure, read-only access for authorized users.

2. Objective:

To gain practical experience in implementing Data Definition Language (DDL), Data Manipulation Language (DML), and Data Control Language (DCL) operations in a real database environment. This will also include implementing role-based privileges to secure data.

3. Experiment Steps:

a) Create table queries:

- i. `CREATE TABLE DEPARTMENT (DEPT_ID INT PRIMARY KEY, DEPT_NAME VARCHAR(20) NOT NULL UNIQUE );`
- ii. `CREATE TABLE PROJECT (PROJ_ID INT PRIMARY KEY, PROJ_NAME VARCHAR(50) NOT NULL, DEPT_ID INT NOT NULL, FOREIGN KEY (DEPT_ID) REFERENCES DEPARTMENT(DEPT_ID) );`
- iii. `CREATE TABLE EMPLOYEE (EMP_ID VARCHAR(10) PRIMARY KEY, EMP_NAME VARCHAR(30) NOT NULL, DEPT_ID INT NOT NULL, PROJ_ID INT NOT NULL, FOREIGN KEY (DEPT_ID) REFERENCES DEPARTMENT (DEPT_ID) ON DELETE CASCADE ON UPDATE CASCADE, FOREIGN KEY (PROJ_ID) REFERENCES PROJECT (PROJ_ID) ON DELETE CASCADE ON UPDATE CASCADE);`

b) Insert queries:

- i. INSERT INTO DEPARTMENT VALUES (1, 'Sales'), (2, 'Marketing'), (3, 'Legal'), (4, 'IT'), (5, 'Accounts'), (6, 'HR');
- ii. INSERT INTO PROJECT VALUES (1000, 'Lead Gen Campaign', 1);
- iii. INSERT INTO EMPLOYEE VALUES ('E100', 'Vikram Singh', 1, 1000);

c) DML UPDATE and DELETE

- i. UPDATE EMPLOYEE SET PROJ_ID = 6000 WHERE PROJ_ID = 6001;
- ii. DELETE FROM PROJECT WHERE DEPT_ID = 3;

d) Access Control and Security

- i. Creating a user: CREATE USER REPORTER WITH PASSWORD 'reporter@123';
- ii. Granting select privileges: GRANT SELECT ON EMPLOYEE, DEPARTMENT, PROJECT TO REPORTER;
- iii. Explicitly revoking create privilege on the DB: REVOKE CREATE ON SCHEMA PUBLIC FROM REPORTER;

e) Schema modification

- i. Alter table: ALTER TABLE EMPLOYEE ADD COLUMN BASE_SAL INT;
- ii. Alter table to remove foreign key constraint further to drop project table: ALTER TABLE EMPLOYEE DROP CONSTRAINT EMPLOYEE_PROJ_ID_FKEY;
- iii. Drop table: DROP TABLE PROJECT;

4. Output

	emp_id [PK] character varying (10)	emp_name character varying (30)	dept_id integer	proj_id integer	base_sal integer
1	E100	Vikram Singh	1	1000	[null]
2	E101	Ritu Das	5	5000	[null]
3	E102	Sonia Patel	6	6000	[null]
4	E103	Kavita Sharma	6	6000	[null]
5	E104	Arjun Gupta	1	1000	[null]
6	E105	Sonia Gupta	1	1000	[null]
7	E107	Neha Gupta	6	6000	[null]
8	E108	Vikram Patel	1	1000	[null]
9	E109	Kavita Gupta	4	4000	[null]
10	E110	Arjun Das	2	2000	[null]
11	E111	Amit Yadav	1	1001	[null]
12	E113	Amit Sharma	6	6000	[null]
13	E114	Vikram Sharma	5	5000	[null]
14	E115	Vikram Kumar	1	1000	[null]
15	E116	Deepak Yadav	6	6000	[null]
16	E117	Kavita Das	2	2000	[null]
17	E118	Sonia Das	1	1001	[null]

5. Learning Outcome

- Learning DDL, DML, and DCL commands in SQL.
- How to implement integrity and domain constraints in SQL.
- DBMS privilege management.